

Homework #5

Channel Capacity

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw5.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw5.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW5.1 *Hamming codes.* Please use the [Hamming code note](#), watch the Hamming code videos below, and solve the following problems.

- [Video 1: Hamming Code](#)
- [Video 2: Hamming Code as a Channel Code](#)
- [Video 3: Hamming Codes—Encoding and Decoding](#)
- [Video 4: Hamming Codes—Erasure Error Correction](#)

(a) Do the set of the first four problems given at the end of the Hamming code note:

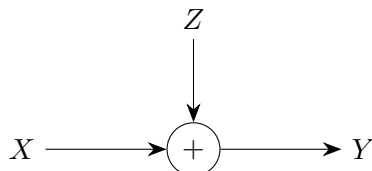
- Find all single error patterns of $(7, 4)$ Hamming codes. Show each error pattern is correctable.
- Find all double error patterns of $(7, 4)$ Hamming codes. Prove/disprove: some double error patterns are correctable.
- Find all single and double erasure patterns of $(7, 4)$ Hamming codes. Prove/disprove: all up to double erasures can be corrected.
- Prove/disprove: there are some triple erasures that can be corrected.

(The last problem of that note—find the probability of information bit error—is difficult, and thus it will be discussed in class.)

- Complete the standard array of the $(7, 4)$ Hamming code.
- What is the capacity of the bit-flip channel? Assume i.i.d. channels.
- What is the rate of the $(7, 4)$ Hamming code?
- What is the length of a codeword? How many channel-uses should you use to send a codeword?
- What is the size of the codebook?
- What is the length of a message?

Problem HW5.2 BSC capacity from the lead $I(X; Y) = H(X) - H(X|Y)$. Prove the capacity of the BSC again with the lead $I(X; Y) = H(X) - H(X | Y)$. (Theorem 5.8 of the Course Book, Capacity of the BSC, used the other expansion, $I(X; Y) = H(Y) - H(Y | X)$; this one requires bounding $H(X | Y)$ from below.)

Problem HW5.3 (C&T 7.2) Additive noise channel. Find the channel capacity of the following discrete memoryless channel:



where $P\{Z = 0\} = P\{Z = a\} = \frac{1}{2}$. The alphabet for x is $\mathbf{X} = \{0, 1\}$. Assume that Z is independent of X . Observe that the channel capacity depends on the value of a .

Problem HW5.4 (C&T 7.3) Channels with memory have higher capacity. Consider a binary symmetric channel with $Y_i = X_i \oplus Z_i$, where $\oplus$ is mod 2 addition, and $X_i, Y_i \in \{0, 1\}$. Suppose that $\{Z_i\}$ has constant marginal probabilities $P\{Z_i = 1\} = p = 1 - P\{Z_i = 0\}$, but that $Z_1, Z_2, \dots, Z_n$ are not necessarily independent. Assume that Z^n is independent of the input X^n . Let $C = 1 - H(p, 1 - p)$. Show that

$$\max_{p(x_1, x_2, \dots, x_n)} I(X_1, X_2, \dots, X_n; Y_1, Y_2, \dots, Y_n) \geq nC.$$

Problem HW5.5 (C&T 7.4) Channel capacity. Consider the discrete memoryless channel $Y = X + Z \pmod{11}$, where

$$Z = \begin{pmatrix} 1, & 2, & 3 \\ \frac{1}{3}, & \frac{1}{3}, & \frac{1}{3} \end{pmatrix}$$

and $X \in \{0, 1, \dots, 10\}$. Assume that Z is independent of X .

- Find the capacity.
- What is the maximizing $p^*(x)$?

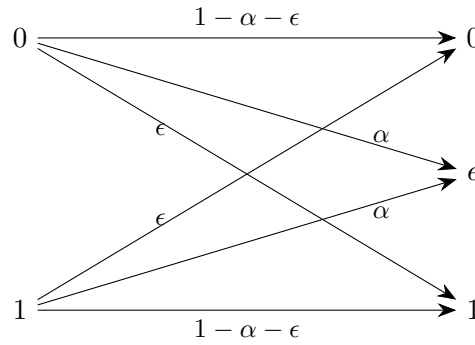
Problem HW5.6 (C&T 7.8) Z-channel. The Z-channel has binary input and output alphabets and transition probabilities $p(y|x)$ given by the following matrix:

$$Q = \begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix} \quad x, y \in \{0, 1\}$$

Find the capacity of the Z-channel and the maximizing input probability distribution.

Problem HW5.7 (C&T 7.9) Suboptimal codes. For the Z-channel of Problem 7.8, assume that we choose a $(2^{nR}, n)$ code at random, where each codeword is a sequence of *fair* coin tosses. This will not achieve capacity. Find the maximum rate R such that the probability of error $P_e^{(n)}$, averaged over the randomly generated codes, tends to zero as the block length n tends to infinity.

Problem HW5.8 (C&T 7.13) Erasures and errors in a binary channel. Consider a channel with binary inputs that has both erasures and errors. Let the probability of error be ϵ and the probability of erasure be α , so the channel is as follows:



- (a) Find the capacity of this channel.
- (b) Specialize to the case of the binary symmetric channel ($\alpha = 0$).
- (c) Specialize to the case of the binary erasure channel ($\epsilon = 0$).

Problem HW5.9 (C&T 7.26) Noisy typewriter. Consider the channel with $x, y \in \{0, 1, 2, 3\}$ and transition probabilities $p(y|x)$ given by the following matrix:

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

- (a) Find the capacity of this channel.
- (b) Define the random variable $z = g(y)$, where

$$g(y) = \begin{cases} A & \text{if } y \in \{0, 1\} \\ B & \text{if } y \in \{2, 3\}. \end{cases}$$

For the following two PMFs for x , compute $I(X; Z)$:

(i)

$$p(x) = \begin{cases} \frac{1}{2} & \text{if } x \in \{1, 3\} \\ 0 & \text{if } x \in \{0, 2\}. \end{cases}$$

(ii)

$$p(x) = \begin{cases} 0 & \text{if } x \in \{1, 3\} \\ \frac{1}{2} & \text{if } x \in \{0, 2\}. \end{cases}$$

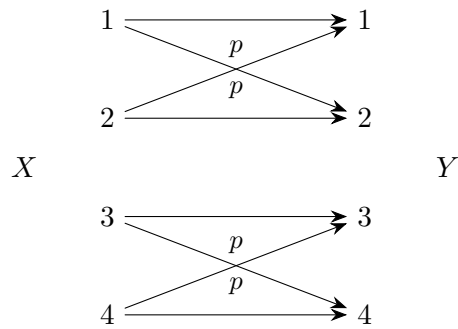
- (c) Find the capacity of the channel between x and z , specifically where $x \in \{0, 1, 2, 3\}$, $z \in \{A, B\}$, and the transition probabilities $P(z|x)$ are given by

$$p(Z = z | X = x) = \sum_{g(y_0)=z} P(Y = y_0 | X = x).$$

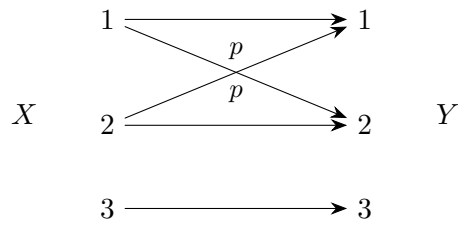
- (d) For the X distribution of part (i) of (b), does $X \rightarrow Z \rightarrow Y$ form a Markov chain?

Problem HW5.10 (C&T 7.34) Capacity. Find the capacity of

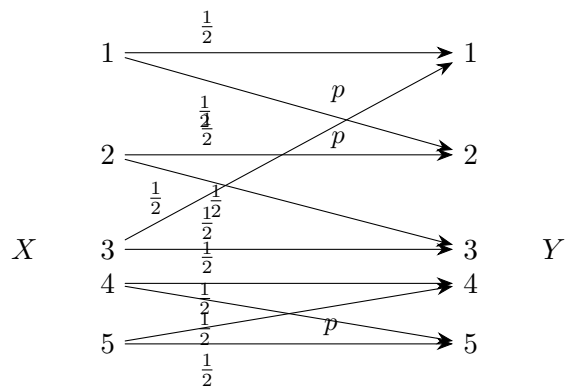
- (a) Two parallel BSCs:



(b) BSC and a single symbol:



(c) BSC and a ternary channel:



(d) Ternary channel:

$$p(y|x) = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix}. \quad (7.167)$$